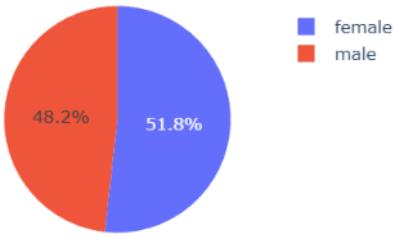
Student Performance Analysis

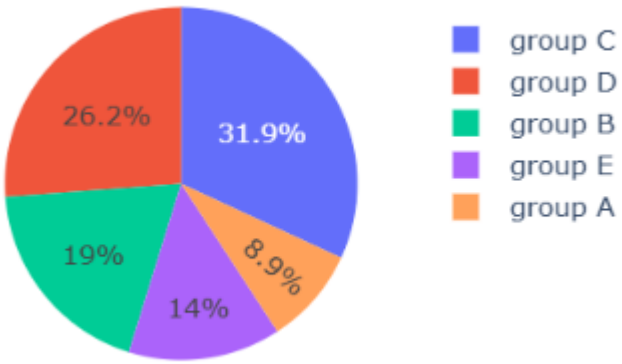
- **Probleam Statement** - This project understands how the students performance(test scores) is effected by other variables such as Gender, Ethnicity, Parental level of education, Lunch and Test preparation course.

- **Columns -** 'gender', 'race/ethnicity', 'parental level of education', 'lunch', 'test preparation course', 'math score', 'reading score', 'writing_score'

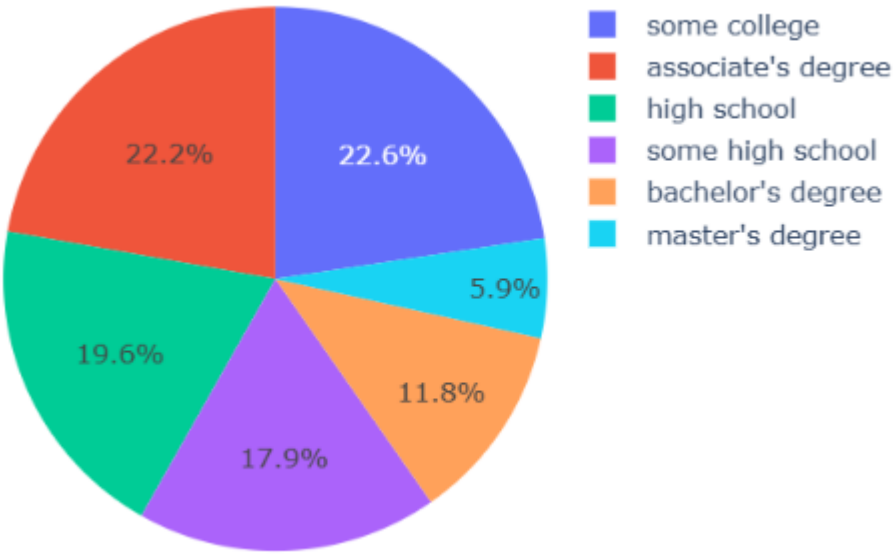
- **Gender:**



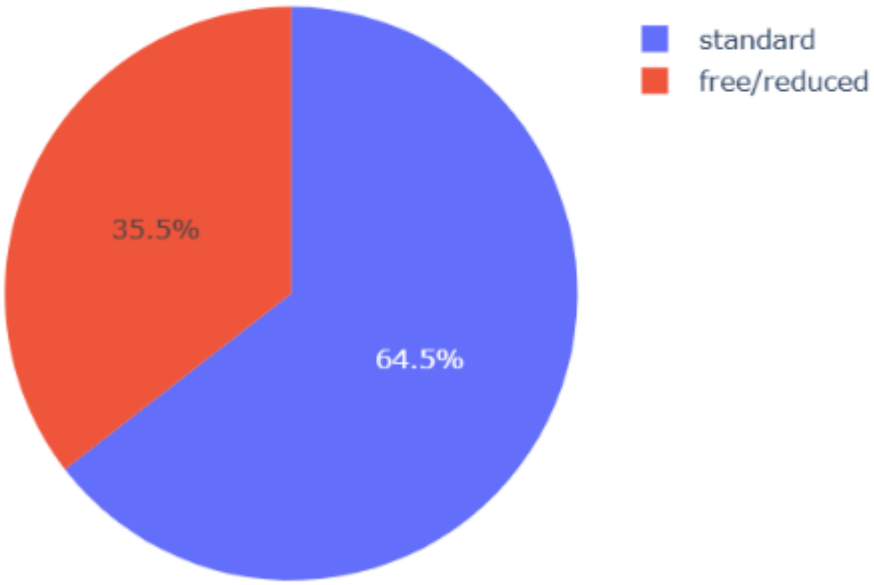
- **Ethnicity:**



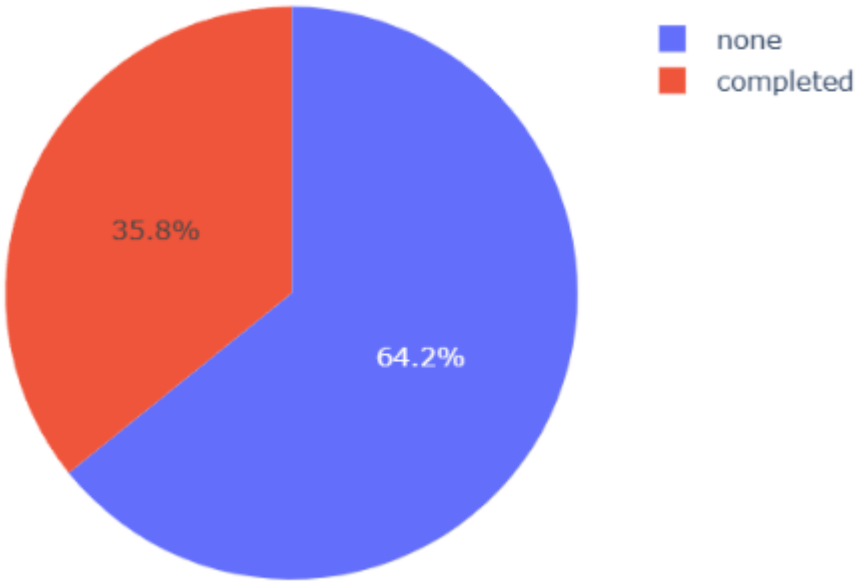
- **Parent Level Of Education:**



- **Lunch**:



- **Test Preparation Course:**



- **Math Score:**
  - this clas hava a good student in maths subject, this data has a left skewed, most studnet have a good mark.
  - but have a min mark is 0 this means some also weak student in class.

mean	66.08900
std	15.16308
min	0.00000
25%	57.00000
50%	66.00000
75%	77.00000
max	100.00000

- **Reading Score:**
  - also good student in reading subject.
  - this data also left skewed and none of the student have a marks means this subject is easy or we have a good student in class

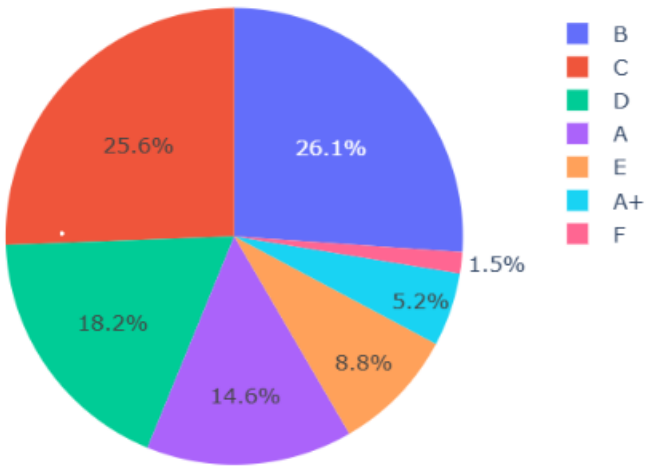
mean	69.169000
std	14.600192
min	17.000000
25%	59.000000
50%	70.000000
75%	79.000000
max	100.000000

- **Writing Score:**
  - also good student in writing subject.
  - this data also left skewed and none of the student have a 0 marks means this subject is easy or we have a good student in class.

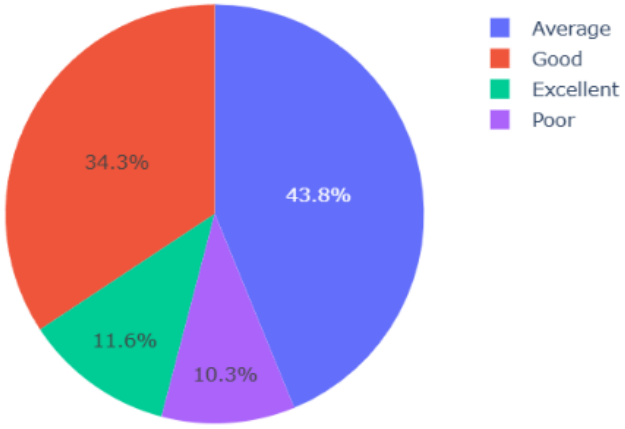
mean	68.054000
std	15.195657
min	10.000000
25%	57.750000
50%	69.000000
75%	79.000000
max	100.000000

- after this we add a Percentage column, and based on this we find a grade of every student.

- **Grade**:

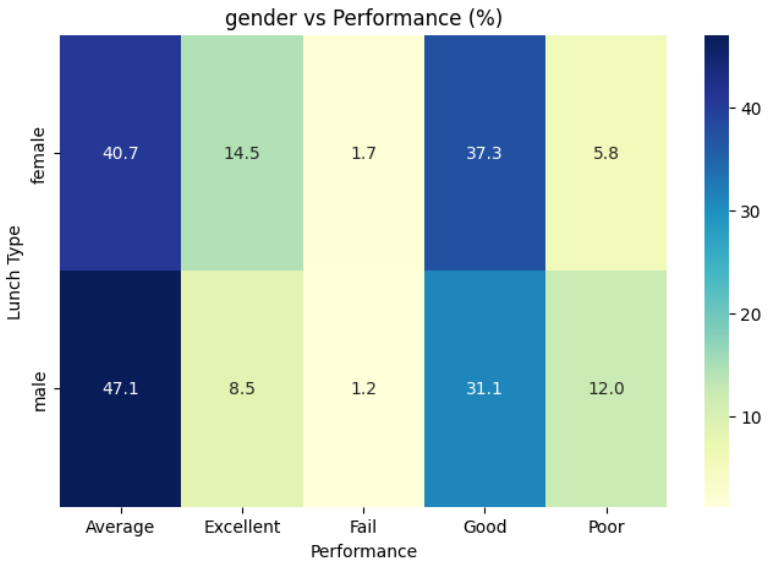


- We add another column performance based on percentage



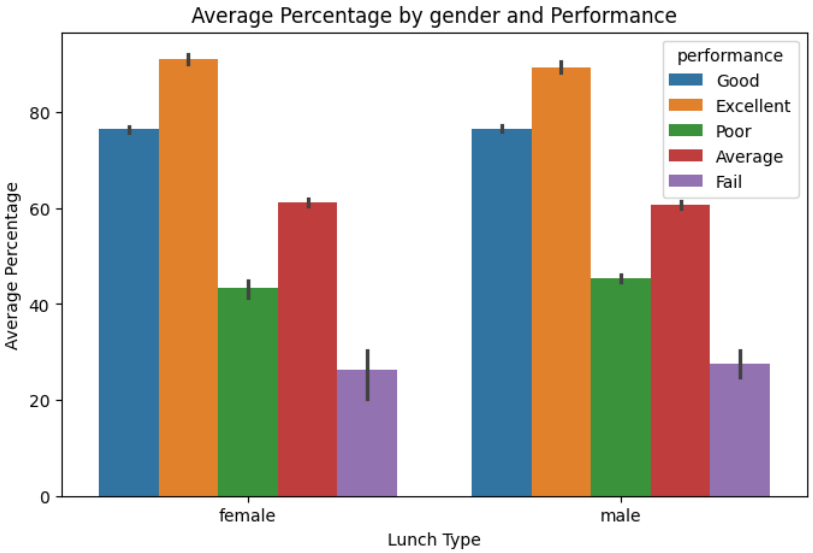
- **Analysis**

1. Gender

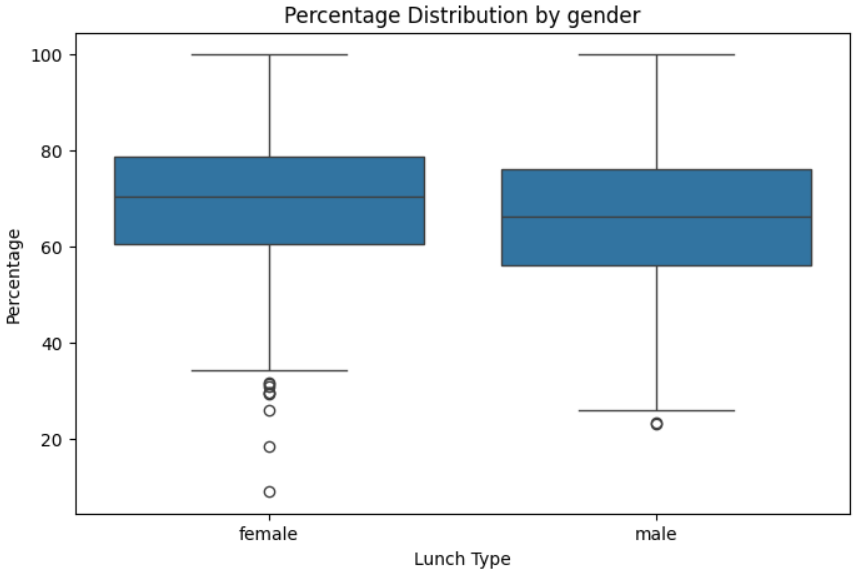


- This chart clearly shows that in this class, **most students fall in the Average category**.
- In **female**, there are **more Excellent students** compared to **male**.
- In **male**, there are **more Poor mark students** compared to **female**.

- In **female**, there are **more Good students** compared to **male**.
- In short, **Good and Excellent students are mostly female**, while **Poor and Average students are mostly male**.

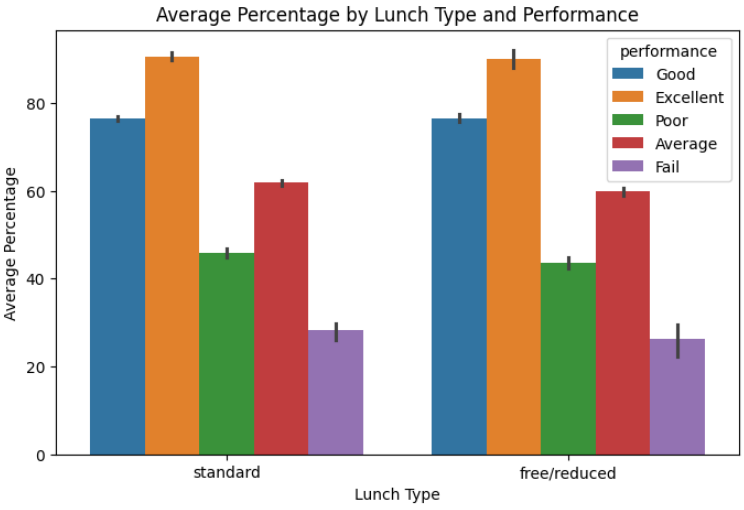


- This is the bar plot of **gender**; overall, there is **no major difference** between **male** and **female** students.
- **Good** students maintain an average percentage of around **80** for **both** genders.
- **Excellent** students maintain an average percentage of around **90** for **both** genders.
- **Poor** students maintain an average percentage of around **40** for **both** genders.
- **Average** students maintain an average percentage of around **60** for **both** genders.
- **Fail** students maintain an average percentage of around **20** for **both** genders.



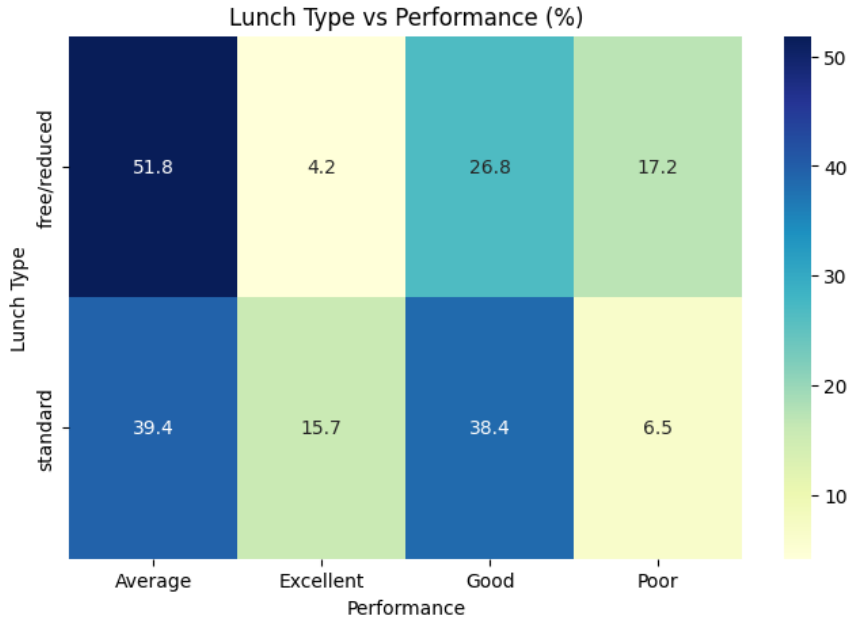
- **Female** student **median percentage** are **more than male** but some **more female student are weak** than male.

2. <u>Lunch</u>:



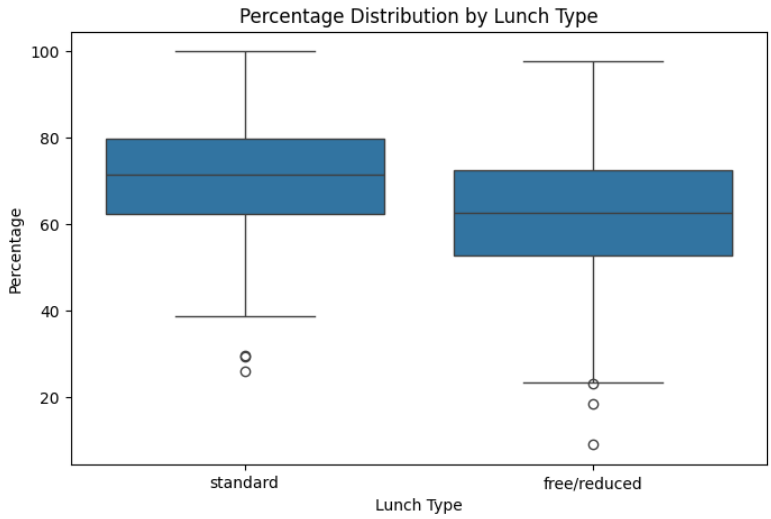
- **Excellent** performers show the **highest average percentage (~90+)** in both lunch types, indicating top performers are equally strong across groups.
- **Good** performers display a **similar average (~75–78)** for both lunch types, with **no significant gap**.

- **Poor** performers in the **free/reduced** group score **lower (~40)** than those in the **standard** group **(~45)**, suggesting weaker students in the free/reduced category perform worse.
- **Average** performers show a **slight edge for standard lunch students (~62)** over free/reduced **(~60)**.
- Overall, **standard lunch students** maintain a **marginally higher average** across most performance bands.



- **free/reduced lunch students** are concentrated in the **Average** category (**51.8%**), indicating most of them perform at an average level.
- **standard lunch students** are more represented in **Excellent (15.7%)** and **Good (38.4%)**, reflecting stronger overall performance.
- The **Excellent** rate for free/reduced is only **4.2%**, compared to **15.7%** for standard — nearly **4× higher** for standard lunch students.
- Among free/reduced students, **Poor** is only **17.2%**, but **Average dominates at 51.8%**, showing they mostly cluster in the middle range.

- For standard lunch students, **Poor** is just **6.5%**, indicating fewer weak performers

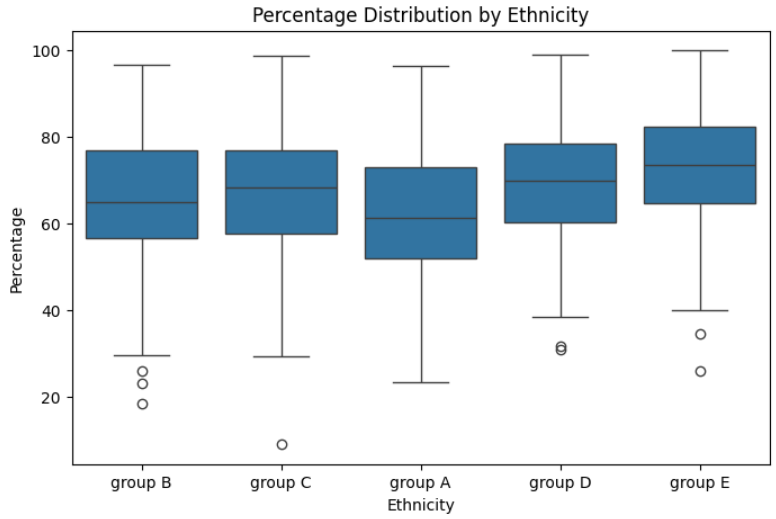


- **Standard lunch students** have a **higher median percentage (~68)** compared to **free/reduced students (~60)**, indicating better typical performance.
- The **interquartile range (IQR)** for **standard lunch** is **wider and positioned higher (~60–80)**, showing a stronger concentration of above-average performers.
- **Free/reduced students** show a **lower IQR (~52–72)** with the median shifted downward, reflecting overall weaker performance.
- **Both groups have outliers** on the lower end; however, **free/reduced students show more extreme low outliers (~10, ~20)**, indicating a few very poor performers.
- The **overall range** is similar for both groups (up to ~100), but the **bulk of free/reduced scores lies lower** than that of standard students.

3. Ethnicity

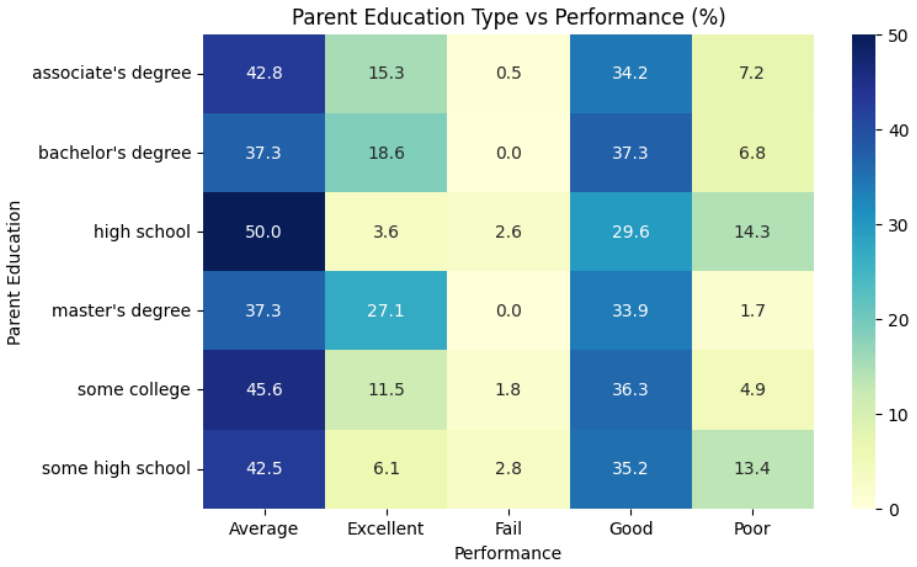


- **group A** has the **highest Average percentage (53.9%)**, showing most of its students cluster in the **Average** band.
- **group E** has the **highest Excellent rate (21.4%)** and **highest Good rate (40.7%)**, indicating it has the **strongest overall performers**.
- **group E** also has the **lowest Average (31.4%)**, meaning fewer of its students are stuck in the middle range.
- **group D** shows a strong **Good performance (39.3%)**, second only to group E, with a **low Fail rate (0.8%)**.
- **group A has the lowest Good (19.1%)** and **highest Average (53.9%)**, reflecting **weaker overall performance** compared to other groups.
- **Fail rates are low across all groups (0.7%–3.2%)**, with **group B having the highest Fail rate (3.2%)**.
- **Poor performance is highest in group A (16.9%)** and lowest in **group E (5.7%)**, confirming the performance gap between these two groups.

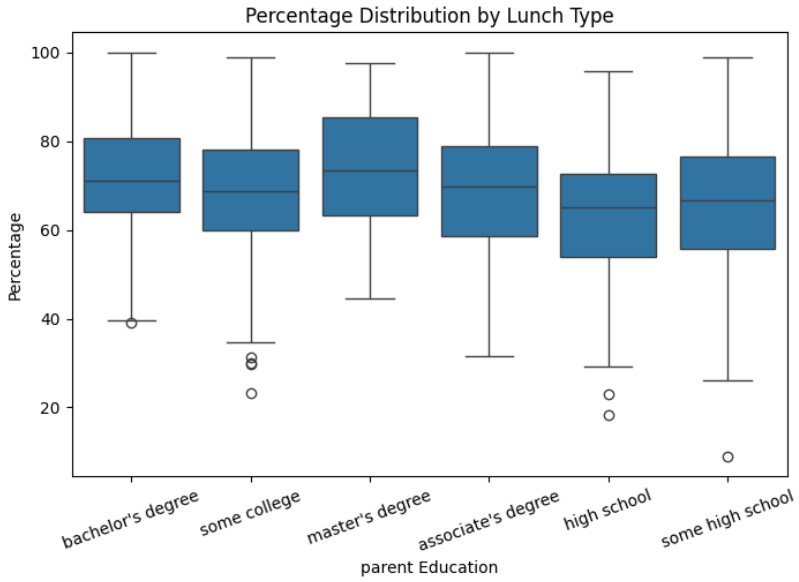


- **group E has the highest median (~74)** and the **highest IQR (~65–82)**, confirming it is the **strongest performing group** — this matches the heatmap finding.
- **group A has the lowest median (~60)** and a **lower IQR (~52–72)**, confirming it is the **weakest group** — this also matches the heatmap.
- **group C has an extreme low outlier (~10)**, which the heatmap did **not reveal** — this is **new information** showing a few very poor performers in group C.
- **group D has visible low outliers (~30, ~32)**, which is **new information** not visible in the heatmap.
- **group E also has low outliers (~28, ~35)**, again **new information** — despite being the strongest group, it still has a few weak students.
- **group B has low outliers (~18, ~22, ~25)**, showing some very low performers — not visible in the heatmap.
- The **spread (IQR width) is similar across all groups**, but the **median shifts upward from group A → group E**, showing a **clear performance gradient**.

4. Parent Education

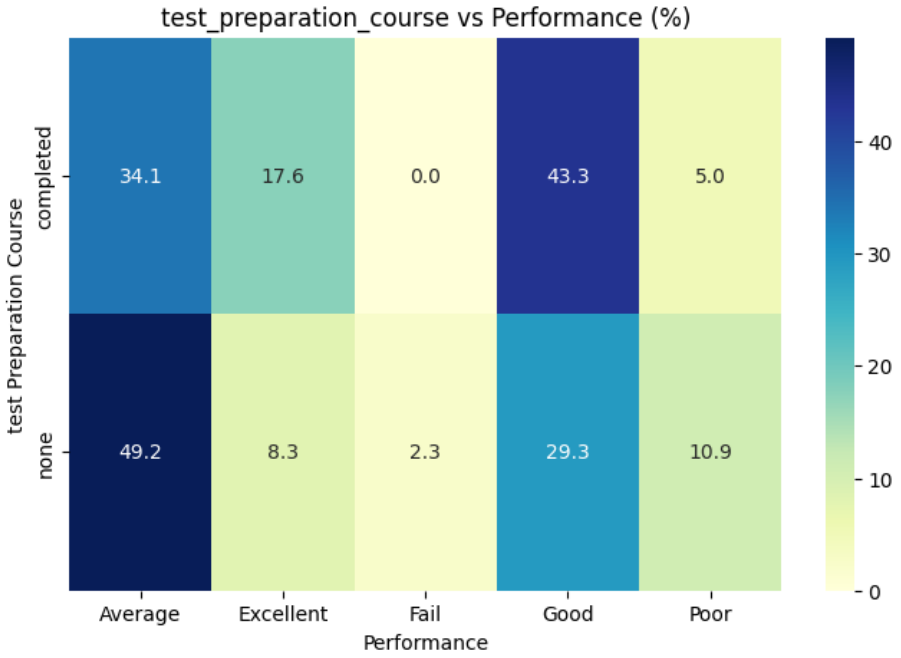


- **master's degree** parents' children have the **highest Excellent rate (27.1%)** and **lowest Poor rate (1.7%)**, indicating **strongest overall performance**.
- **bachelor's degree** group shows a **balanced profile** — **Excellent (18.6%)** and **Good (37.3%)**, with a **0.0% Fail rate**.
- **high school** group has the **highest Average (50.0%)** and **highest Poor (14.3%)**, reflecting **weaker overall performance**.
- **some high school** group also shows **high Poor (13.4%)** and **low Excellent (6.1%)**, indicating **weak performance**.
- **associate's degree** group has a **high Average (42.8%)** with a **modest Excellent (15.3%)**, showing **average-level performance**.
- **some college** group performs **moderately** with **Good (36.3%)** and **Average (45.6%)**, but a **low Excellent (11.5%)**.
- **Fail rates are low across all groups (0.0%–2.8%)**, with **high school having the highest (2.6%)**.

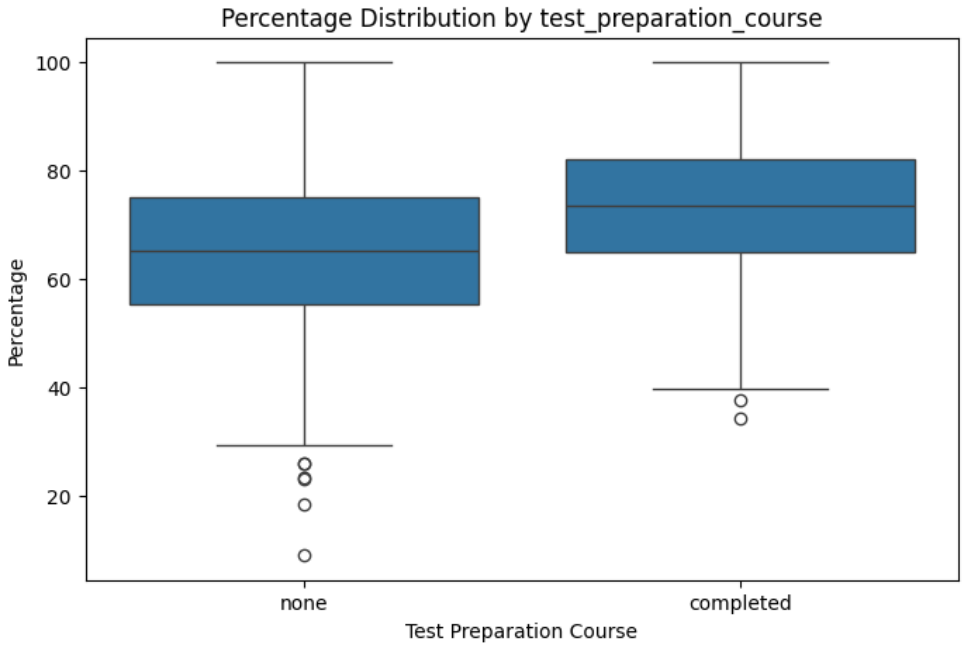


- **group C has an extreme low outlier (~10)** — the **lowest single score** across all groups, not visible in the heatmap.
- **group E has low outliers (~28, ~35)** despite being the **top-performing group**, showing it still contains a few weak students.
- **group D shows low outliers (~30, ~32)**, indicating some underperformers hidden within its strong average.
- **group B has multiple low outliers (~18, ~22, ~25)**, revealing a small cluster of very weak students.
- **group A has the lowest median (~60) and no extreme outliers**, meaning its weak performance is **consistent**, not driven by a few students.

5. Test Preparation Course



- **completed** students have a **higher Excellent rate (17.6%)** and **Good rate (43.3%)**, showing **stronger overall performance**.
- **none** students have a **higher Average rate (49.2%)** and **higher Poor rate (10.9%)**, indicating **weaker performance**.
- **completed** students have a **0.0% Fail rate**, while **none** students have **2.3% Fail rate** — a **clear advantage** for course completers.
- The **Good category** is the biggest difference: **completed (43.3%)** vs **none (29.3%)** — nearly **1.5× higher**.
- **Excellent rate** for **completed (17.6%)** is **more than double** that of **none (8.3%)**.

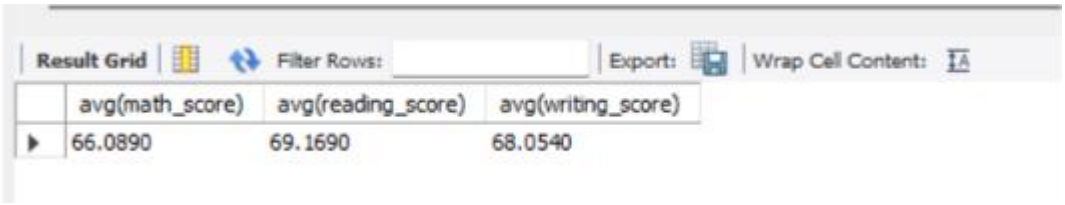


- **completed** students have a **higher median (~72)** compared to **none (~64)**, confirming better typical performance.
- The **IQR for completed (~63–82)** is **positioned higher and slightly narrower** than **none (~55–75)**, showing **more consistent high performance**.
- **none** students show **multiple low outliers (~10, ~22, ~25)**, including the **lowest single score (~10)** in the dataset.
- **completed** students have **only two mild outliers (~35, ~38)**, meaning **very few weak performers**.
- The **lower whisker for none extends to ~30**, while for **completed it stops at ~40**, showing **completed students rarely fall below 40%**.

- **Now we upload this data on mysql server and solve some question**

1. Calculate the average marks of Math, Reading, and Writing, Identify which subject has the highest average score.

```
select avg(math_score), avg(reading_score),
avg(writing_score)
from student;
```

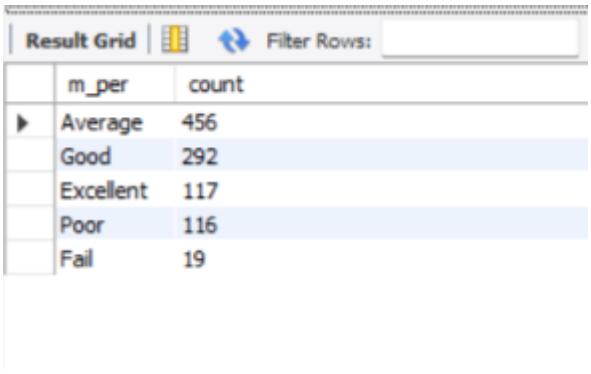


Result Grid | Filter Rows: | Export: | Wrap Cell Content:

avg(math_score)	avg(reading_score)	avg(writing_score)
66.0890	69.1690	68.0540

2. Based on m_per, how many students are in each Math performance category (Fail, Poor, Average, Good, Excellent)?

```
select m_per, count(*) as 'count'
from student
group by m_per
order by count desc;
```

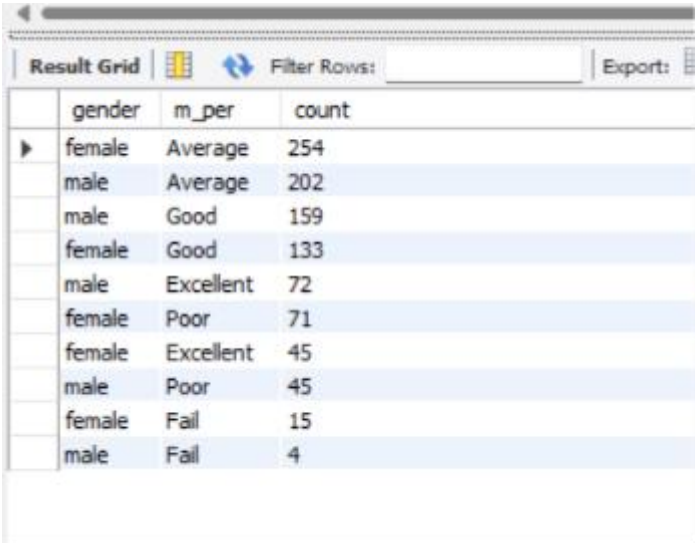


Result Grid | Filter Rows:

m_per	count
Average	456
Good	292
Excellent	117
Poor	116
Fail	19

4. For each gender, count the number of students in each Math performance category.

```
select gender, m_per,
count(*) as 'count'
from student
group by gender, m_per
order by count desc;
```

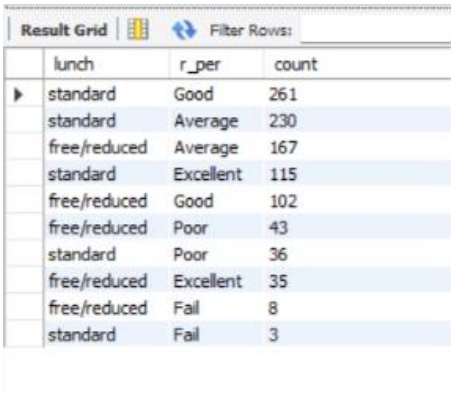


Result Grid | Filter Rows: | Export:

gender	m_per	count
female	Average	254
male	Average	202
male	Good	159
female	Good	133
male	Excellent	72
female	Poor	71
female	Excellent	45
male	Poor	45
female	Fail	15
male	Fail	4

5. **Based on lunch, count the students in each r_per category.**

```
select  lunch, r_per,
count(*) as 'count'
from student
group by lunch, r_per
order by count desc;
```



Result Grid | Filter Rows:

lunch	r_per	count
standard	Good	261
standard	Average	230
free/reduced	Average	167
standard	Excellent	115
free/reduced	Good	102
free/reduced	Poor	43
standard	Poor	36
free/reduced	Excellent	35
free/reduced	Fail	8
standard	Fail	3

6. **Compare the average math_score, reading_score, and writing_score of male and female students.**

```
select gender,
avg(math_score),
avg(reading_score),
avg(reading_score)
from student
group by gender;
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

gender	avg(math_score)	avg(reading_score)	avg(reading_score)
female	63.6332	72.6081	72.6081
male	68.7282	65.4730	65.4730

7. **Calculate the average scores of Math, Reading, and Writing for each ethnicity group. Identify the highest-average group in each subject.**

```
select ethnicity,
avg(math_score),
avg(reading_score),
avg(writing_score)
from student
group by ethnicity
order by avg(math_score) desc,
avg(reading_score) desc, avg(writing_score) desc
```

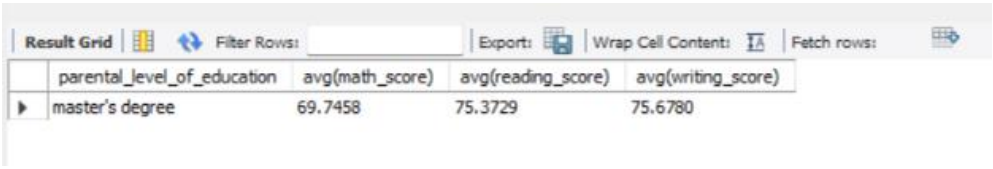


Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ethnicity	avg(math_score)	avg(reading_score)	avg(writing_score)
group E	73.8214	73.0286	71.4071
group D	67.3626	70.0305	70.1450
group C	64.4639	69.1034	67.8276
group B	63.4526	67.3526	65.6000
group A	61.6292	64.6742	62.6742

8. **Calculate the average scores of all three subjects based on parental_level_of_education. Check which parental education category has the highest average in which subject.**

```
select parental_level_of_education,
avg(math_score),
avg(reading_score),
avg(writing_score)
from student
group by parental_level_of_education
order by avg(math_score) desc,
avg(reading_score) desc, avg(writing_score) desc limit 1;
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

parental_level_of_education	avg(math_score)	avg(reading_score)	avg(writing_score)
master's degree	69.7458	75.3729	75.6780

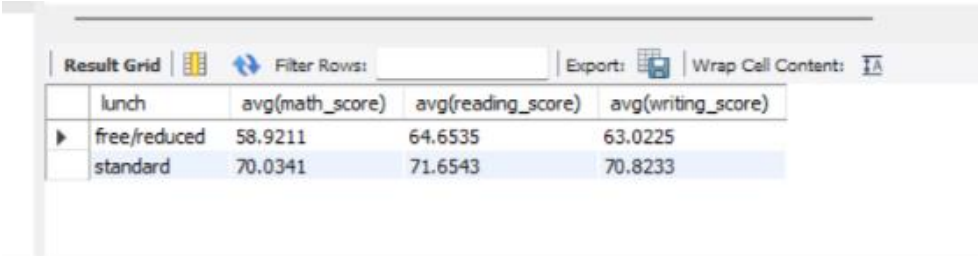
9. **Compare the average Math, Reading, and Writing scores of Standard and free/reduced lunch groups.**

```
select lunch,
avg(math_score),
avg(reading_score),
```

avg(writing_score)

from student

group by lunch



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

lunch	avg(math_score)	avg(reading_score)	avg(writing_score)
free/reduced	58.9211	64.6535	63.0225
standard	70.0341	71.6543	70.8233

10. Compare the average scores of students who completed the course and those who did not. Also calculate the difference in average score for each subject.

select test_preparation_course,

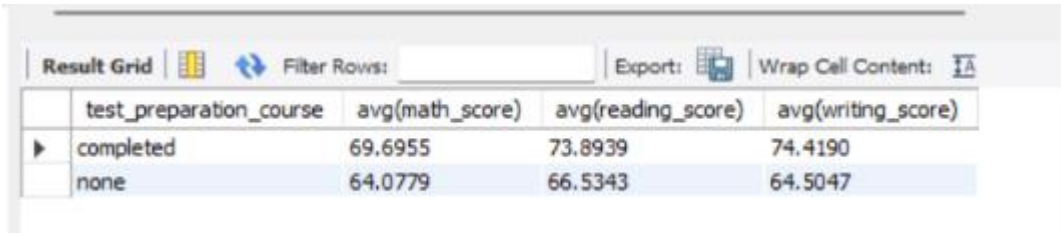
avg(math_score),

avg(reading_score),

avg(writing_score)

from student

group by test_preparation_course;



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

test_preparation_course	avg(math_score)	avg(reading_score)	avg(writing_score)
completed	69.6955	73.8939	74.4190
none	64.0779	66.5343	64.5047

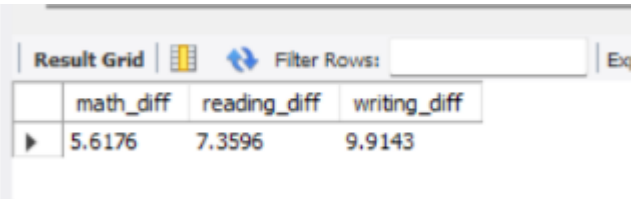
select

avg(case when test_preparation_course = 'completed' then math_score end) -

avg(case when test_preparation_course = 'none' then math_score end) as math_diff,

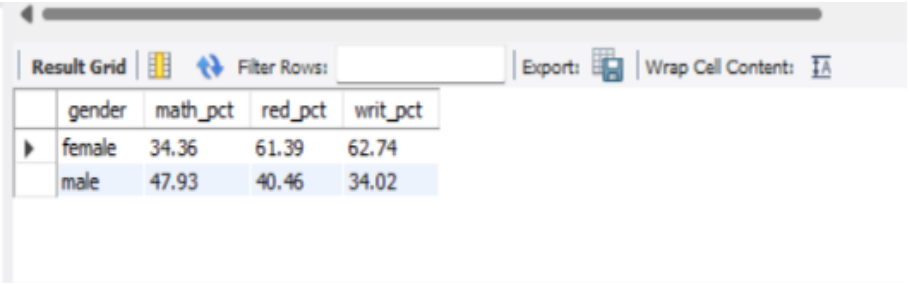
```
  avg(case when test_preparation_course = 'completed' then
reading_score end) -
  avg(case when test_preparation_course = 'none' then
reading_score end) as reading_diff,

  avg(case when test_preparation_course = 'completed' then
writing_score end) -
  avg(case when test_preparation_course = 'none' then
writing_score end) as writing_diff
from student;
```



11. What percent of female students are in the Good or Excellent category in m_per?

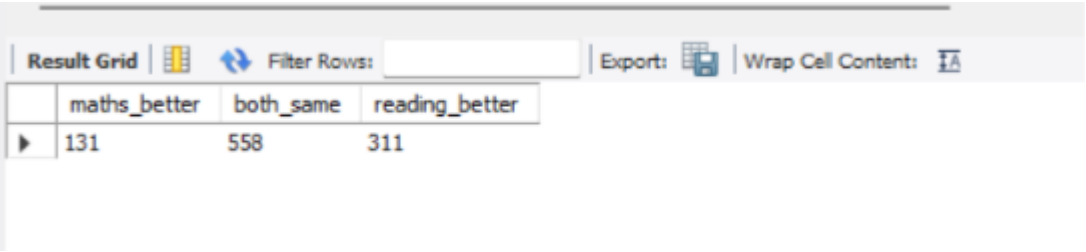
```
select
        gender,
        round(sum(case when m_per in ('good', 'excellent') then 1
else 0 end) / count(*) * 100, 2) as math_pct,
        round(sum(case when r_per in ('good', 'excellent') then 1
else 0 end) / count(*) * 100, 2) as red_pct,
        round(sum(case when w_per in ('good', 'excellent') then 1
else 0 end) / count(*) * 100, 2) as writ_pct
from student
group by gender;
```



gender	math_pct	red_pct	writ_pct
female	34.36	61.39	62.74
male	47.93	40.46	34.02

12. Count the students whose Math performance is better than Reading, both subjects are in the same category, or Reading performance is better than Math.

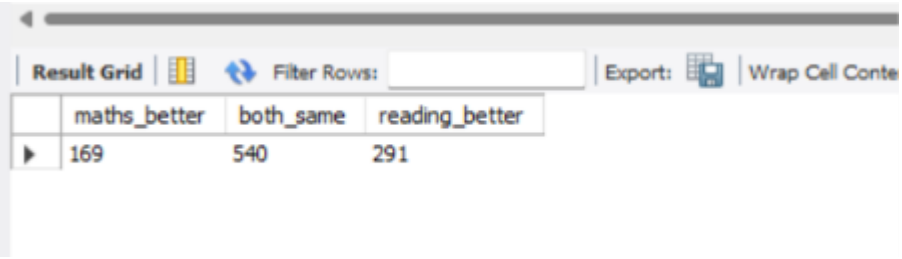
```
select
        sum(case when math_rank > reading_rank then 1 else 0 end) as maths_better,
   sum(case when reading_rank = math_rank then 1 else 0 end) as both_same,
   sum(case when reading_rank > math_rank then 1 else 0 end) as reading_better
from(
select
   case m_per
    when 'fail' then 1 when 'poor' then 2 when 'average' then 3
    when 'good' then 4 when 'excellent' then 5
   end as math_rank,
        case r_per
    when 'fail' then 1 when 'poor' then 2 when 'average' then 3
    when 'good' then 4 when 'excellent' then 5
   end as reading_rank
from student
) as t;
```



maths_better	both_same	reading_better
131	558	311

13. Compare Math and Writing like Q12. Find how many students are stronger in Math and how many in Writing.

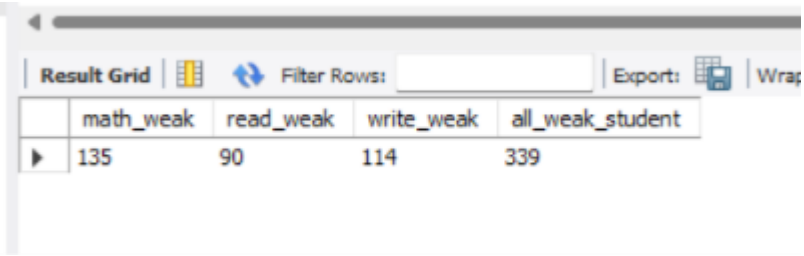
```
select
        sum(case when math_rank > writing_rank then 1 else 0 end) as maths_better,
   sum(case when writing_rank = math_rank then 1 else 0 end) as both_same,
   sum(case when writing_rank > math_rank then 1 else 0 end) as reading_better
from(
select
   case m_per
    when 'fail' then 1 when 'poor' then 2 when 'average' then 3
    when 'good' then 4 when 'excellent' then 5
   end as math_rank,
        case w_per
    when 'fail' then 1 when 'poor' then 2 when 'average' then 3
    when 'good' then 4 when 'excellent' then 5
   end as writing_rank
from student
) as t;
```



maths_better	both_same	reading_better
169	540	291

14.Count the students who are in the Fail or Poor category in all three subjects — Math, Reading, and Writing.

```
select
        math_weak,
   read_weak,
   write_weak,
        math_weak + read_weak + write_weak as all_weak_student
from (
select
        sum(case when m_per in ('fail', 'poor') then 1 else 0 end) as math_weak,
   sum(case when r_per in ('fail', 'poor') then 1 else 0 end) as read_weak,
   sum(case when w_per in ('fail', 'poor') then 1 else 0 end) as write_weak
from student
) t;
```



math_weak	read_weak	write_weak	all_weak_student
135	90	114	339